Avni Infrastructure Overview

This document outlines the infrastructure, security and privacy features of Avni.
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About Avni

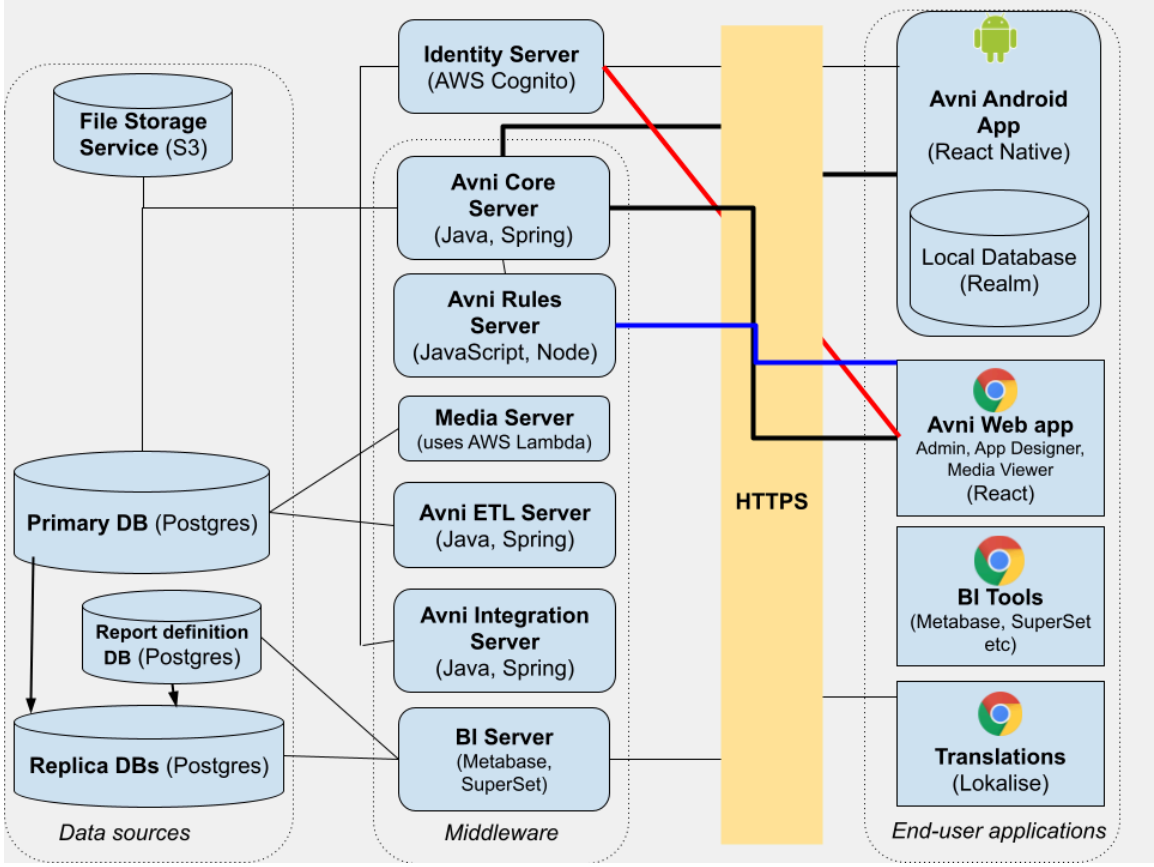
Avni is an open-source platform built for community service and data collection. It has been recognised as [Digital Public Good by the UN Digital Public Goods Alliance]. Key features of Avni include:

1. **Offline**: Avni has an Android application that works completely offline with two-way sync support. When in -network, the users can sync - which pulls configuration and data from the server on cloud and pushes data captured locally.
2. **Rich Case Modelling**: Avni supports creating a custom data model for case management and longitudinal tracking using entities like subject types, programs and encounters. This will help in easily creating the required framework for setting a reporting tool.
3. **Powerful forms**: The data is collected using easily configurable forms. The forms support all standard data types including GPS location, image, video and audio.
4. **Multilingual**: The app can be developed in English and then easily converted into local languages like Assamese by uploading translations. Both android and the web based data entry application support entering and viewing data in regional language. User settings are provided to set the default language.
5. **Location Hierarchy**: We can easily setup location hierarchy and record data against this. E.g. State -> District -> Village -> Household
6. **Users and Catchment**: Users can be easily created in the system and mapped to different geographical regions so that data only limited to that geographical region is available in the android application. E.g. A frontline worker can be mapped to a single village and supervisors can be mapped to block or district.
7. **Bulk data upload and export**: Avni supports downloading data in longitudinal format. This allows the users to extract raw data to perform analysis using tools like Excel, SPSS, Strata or R. Avni also has an ability to upload bulk data using CSV files.
8. **Security**: All the access is behind login screens. All the data is stored on encrypted disks. Users can be set up with Roles and Permissions.
9. **Features for better coordination and engagement with Users:** Avni provides features like Approval Workflow, News broadcast and an issue resolution framework to better engagement with users, coordinate the work and ensure data quality.
10. **Health Specific Features** like Vaccination Tracking and Growth Chart plotting in the health worker mobile application.

Component architecture

The Avni suite has the following components

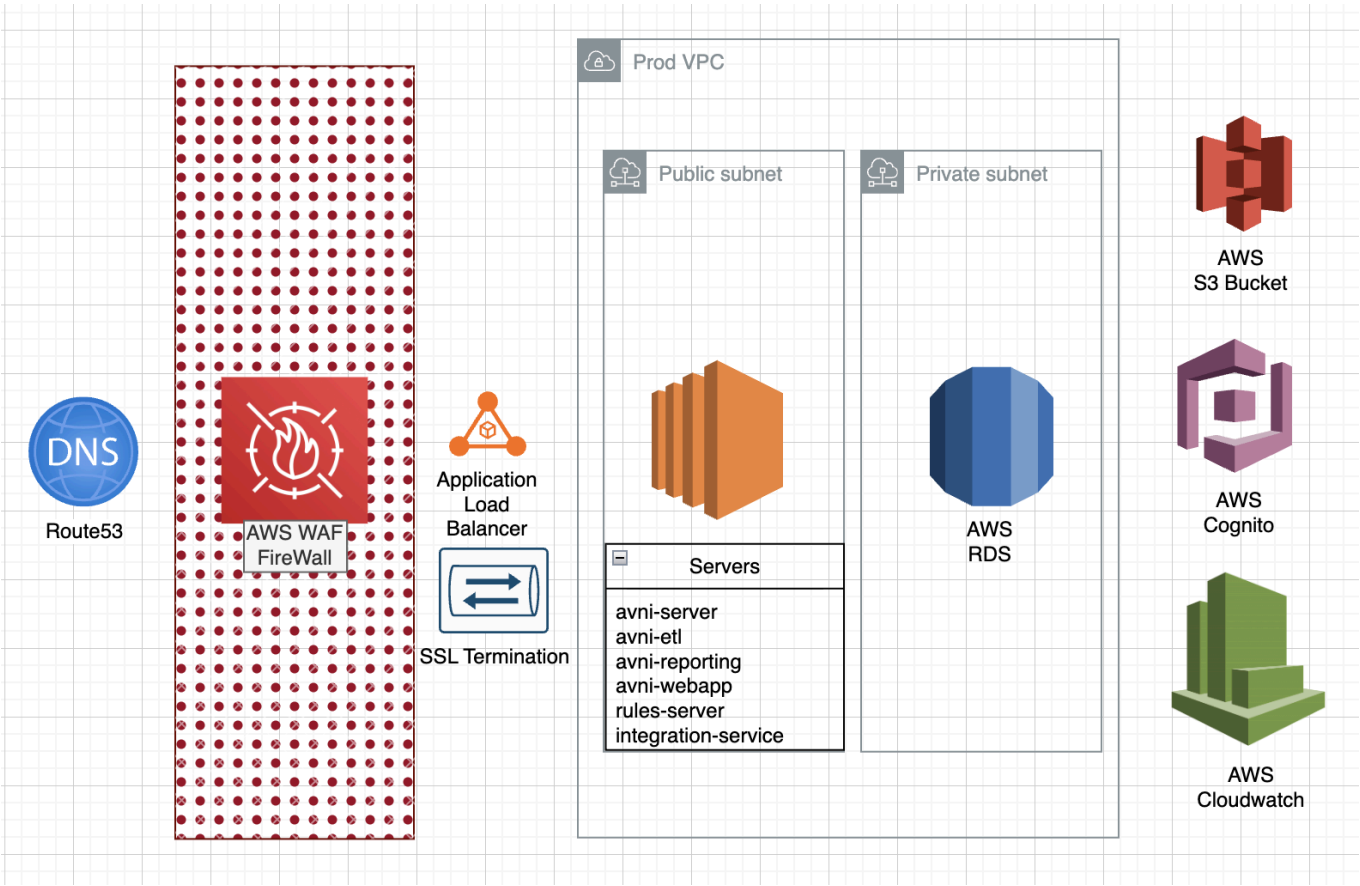
- **Field app (Android)** - Avni app deployed on the Android PlayStore. All field users work on the field app.
- **Administration app and App Designer (Web)** - Web application hosted on app.avniproject.org - This is used by administrators and app designers. Primary purposes include csv uploads, creation of users/locations, and changing form definitions.
- **Avni server** - Primary API server hosted on app.avniproject.org.
- **Rules Server** - Meant to run Javascript rules that are built for the application
- **Media Server** - Handles the Media search mechanism on Avni
- **ETL Server** - ETL service. Transforms primary data to formats easier for reporting
- **Integration Server** - Takes care of integration with external systems
- **BI Server** - An instance of Metabase and another of Superset. These provide dashboards



Notes

- The Android and web applications interact with the server through API interfaces.
- Identity is established through jwt tokens provided by the Identity Server (Amazon Cognito).
- All non-media is stored in the postgres database. The database also stores report definitions for the reports and denormalized tables for use by reports.
- Media is stored in S3

Network Architecture

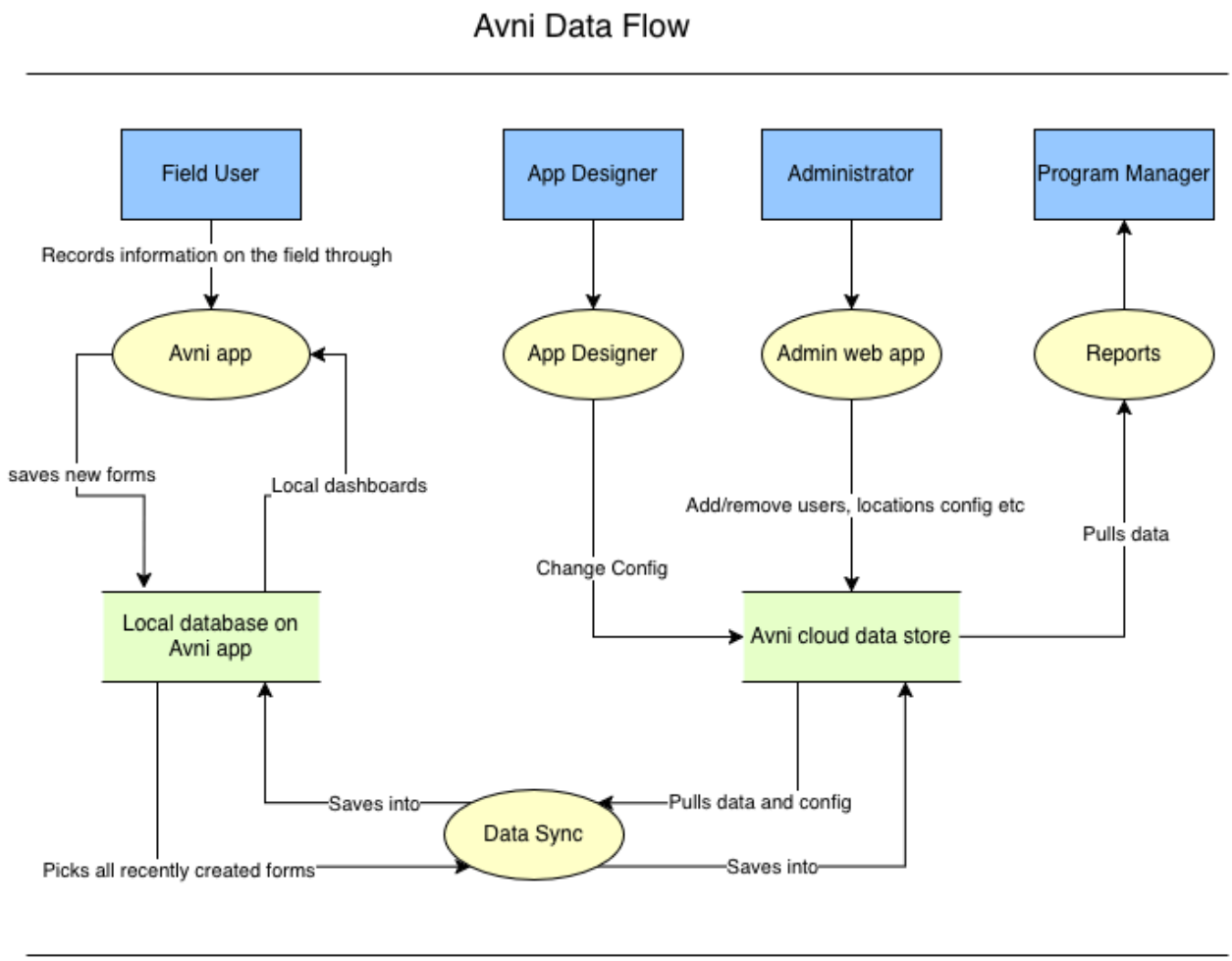


Notes

- The infra is hosted on AWS Mumbai servers
- WAF protects against DDoS attacks
- SSL termination takes place at the Application Load Balancer before being routed to the right server.
- Infra capacity is provided along with the components in the diagram
- All EC2 servers run Ubuntu

- All services run on systemd as services
- Deployment happens through CircleCI

Data Flow



External sdks

Name	Vendor	Purpose	Data handled	Service Terms
Cognito	AWS	Identity Server	Identity of user (user id, password, phone number)	https://aws.amazon.com/service-terms/
APM	NewRelic	Server monitoring	Code instrumentation (execution stack,	https://newrelic.com/termsandconditions/terms

			server response times, base urls)	
Bugsnag	SmartBear	Error management	Stack traces, username, phone characteristics such as version, memory etc	https://smartbear.com/terms-of-use/
Firebase	Google	Analytics	Performance of page views, events	https://marketingplatform.google.com/about/analytics/terms/us/

Notes:

- All sdks that send data outside the data center are provided here
- No information about beneficiaries is shared with any of the tools mentioned above

Security measures

Summary

Area	Item	Implementation Details
Data Security	Encryption at Rest (mobile app)	AES-256 (needs to be enabled for an organisation)
	Encryption at Rest (server)	AES-256 Encrypted by AWS
	Encryption at Rest (backups)	AES-256 Encrypted by AWS
	Encryption of data in transit (public network)	TLS_ECDHE_RSA_WITH_AES_128_GCM_SHA256, 128 bit keys, TLS 1.2
	Data in transit (within VPC)	Trusted
	Database backups	Point in Time recovery for 35 days for the database
	Data residence	AWS Mumbai data centers

App Security	Login	Open ID Connect (OIDC) using Amazon Cognito. JWTs for authentication everywhere
	Token expiry	1 hour
	Role based Access Control	Present. Can be modified by admins.
	Upgrades	Regular upgrades to application packages
Infrastructure	Infrastructure access	MFA mandatory. Access logged. Ssh access through keys (nopasswd)
	Access logging	AWS Cloudwatch
Network Security	Firewall	WAF enabled
	Databases	Within private subnet
	Routing through subnets	Implemented via AWS Security Groups
Deployment	Secrets management for server	Encrypted config deployed through CI servers. Applications read through environment variables
	Secrets management for mobile app	No secrets maintained in the app bundle
Contracts	Customers	NDA added to all contracts
	Employees	NDA added to all contracts
Audits	Security Audit	On demand. Latest report performed for a dedicated deployment Android - Leadership for Equity_ Tea… Webapp - Leadership for Equity_ AVN…

Avni Platform : Data Handling - Access and Sharing of Data

1. **Authentication:** Data in all the applications is accessible only to authenticated users. It is the user's responsibility to ensure that they are keeping a strong password so that it cannot be guessed by another human or brute force attack.
2. **Protection of database:** The database resides within the internal VPC of AWS, and is not publicly accessible. The only way to access the database is through the interfaces provided by Avni, namely the Android application, the web application and the reporting systems. It is protected against accidental deletion, and has PITR (Point in Time Restore) mechanism to restore to any point of time. This protects us from disasters and data corruption.
3. **Authorization / Access Controls:** a. Avni provides the ability to restrict access to functions to specific users through user roles and detailed privileges. b. Only relevant data is made available to the users for their catchments.
4. **Encryption in transit:** Applications when communicating with each other use encryption via HTTPS protocol.
5. **Vulnerabilities from Libraries:** All the libraries and certificates are regularly upgraded to prevent new vulnerabilities.
6. **Protection of PIIs:** There is no additional protection of PII or any other sensitive information once downloaded from the system in the form of spreadsheets by an authenticated and authorised user using reports.

Data Storage [Data at Rest]

1. **Security and encryption of data at Rest:** a). The android field application has an option to store data in an encrypted form. We also rely on android security measures to ensure that other apps don’t have non-authenticated access to Avni’s data. b). Samanvay cloud service uses Amazon cloud storage (AWS) for storage of data on the cloud. Samanvay uses Amazon RDS encryption to secure the DB instances and snapshots at rest. Amazon RDS encryption uses the industry standard AES-256 encryption algorithm to encrypt the data on the server that hosts the DB instance. AWS RDS encryption also ensures that Amazon doesn’t have access to the database and its contents.
2. **Data within the country boundaries:** Samanvay uses the Amazon data centres from India (Mumbai AZ) to store the data.
3. **Access for Troubleshooting purposes:** Authorised Samanvay team members have access to data for troubleshooting purposes. Samanvay and its members will be bound by Non disclosure agreement between NGO partner and Samanvay.

Privacy

Samanvay performs the role of a **Data Processor** as provided in the Digital Personal Data Protection Act, 2023. As part of this, we ensure that the data is secure, accessible only to authorised users, and any requests for PII removal are done on request of our partner. We also provide notice on breach of data within the platform.

We expect our partners to ensure management of consent, data minimisation, and route any request to delete data on behalf of the data principal.